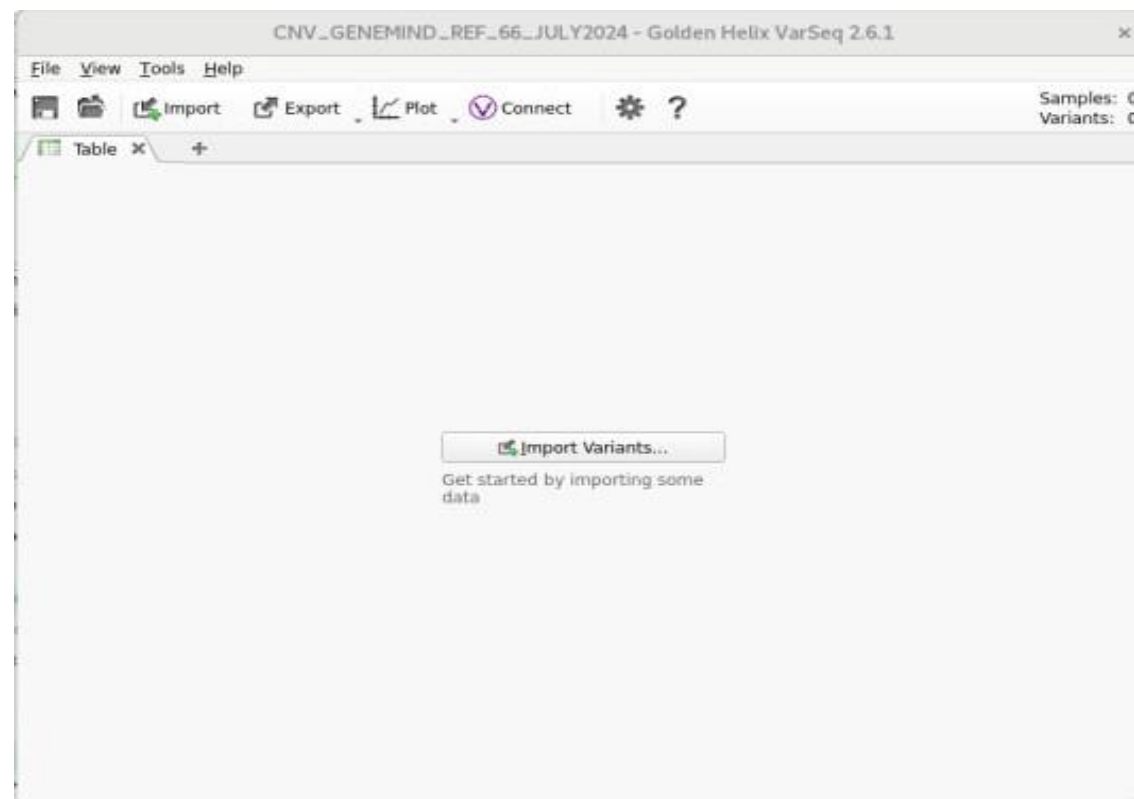
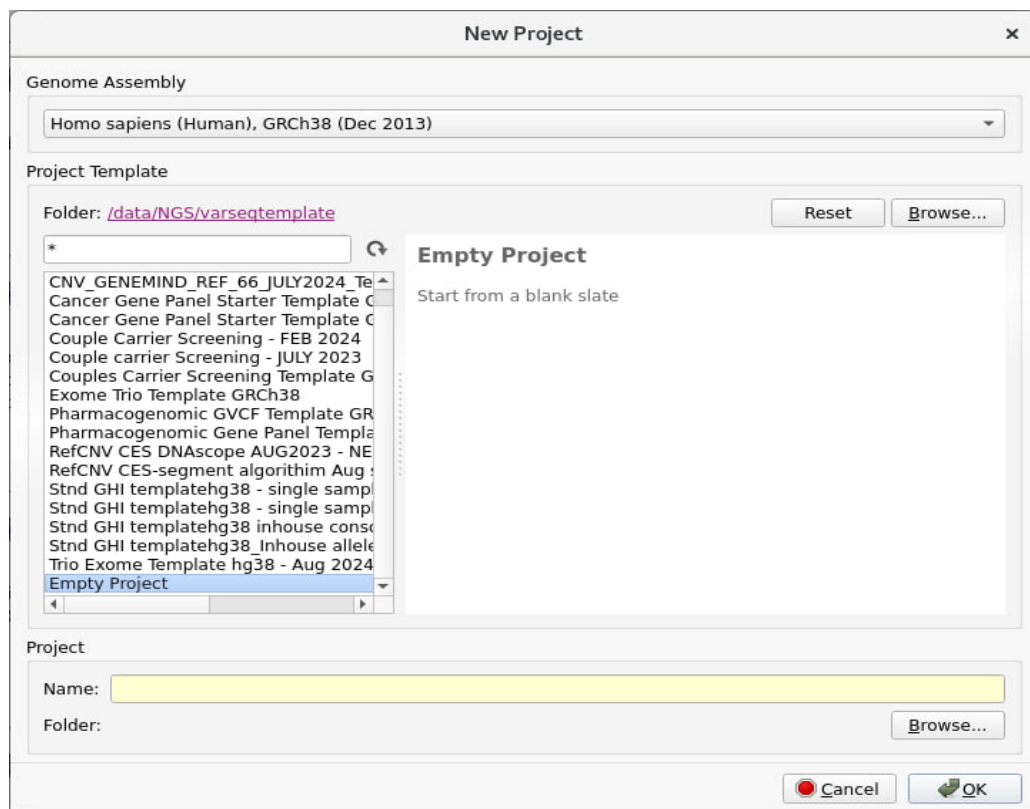


# **CNV OPTIMISATION - SOP**

# STARTING WITH REFERENCE BAM FILE CREATION

Create a folder with VCF and BAM. Remove the VCF that has a true CNV call.



Start with empty project, create a name and locate it in the folder that contains VCF and BAM for CNV

Import Variants Wizard

### Import Variant Sources

- Define Input
- Scan Input
- Change Options
- Review

Select one or more variant files to import.

Select Files:

Add Files  
Remove  
Add Folder  
Remove All

Sample Name Matching Mode

☒ Import Without Appending

☐ Append Samples with the Same Names

☐ Append Samples with Similar Names (Shared Prefixes)

Import Tables

☒ Variant Only

☐ Variants, CNVs

☐ Variants, CNVs, Breakends

☐ Advanced Options

Help

< Back Next > Cancel

DON'T CHANGE IT

Import Variants Wizard

### Import Variant Sources

- Define Input
- Scan Input
- Change Options
- Review

Select the samples of interest and appropriately adjust their attributes

☒ Individual Samples ☐ Family Samples ☐ Tumor/Normal Samples ☐ Partnered Samples

Add sample fields: From Text File Associate Alignment File New Manual Field

Original Samples

|     |                                                                                |
|-----|--------------------------------------------------------------------------------|
| ✓ 1 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_10_GEM_T_10_L |
| ✓ 2 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_11_GEM_T_11_L |
| ✓ 3 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_12_GEM_T_12_L |
| ✓ 4 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_13_GEM_T_13_L |
| ✓ 5 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_14_GEM_T_14_L |
| ✓ 6 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_15_GEM_T_15_L |
| ✓ 7 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_16_GEM_T_16_L |
| ✓ 8 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_1_GEM_T_1_L00 |
| ✓ 9 | 202403071719_AB01231201002_2P240102010U5252699DX_B_DemoWES_GEM_T_2_GEM_T_2_L00 |

☐ Advanced Options

Change the sample names: Set to File Name Set to File Name\_Sample Reset

Help

< Back Next > Cancel

Select Alignment Files to be Associated with Each Sample

Select each sample's corresponding alignment file by selecting it from the drop down menu.

RV\_Raw\_data/GENEMIND/CNV\_VCF AND BAM  Found 66 alignment files

| Sample Name                                                       | File Name |
|-------------------------------------------------------------------|-----------|
| 2024030... B DemoWES_GEM_T_10_GEM_T_10_L00_outfastp_R_deduped.bam |           |
| 2024030... B DemoWES_GEM_T_11_GEM_T_11_L00_outfastp_R_deduped.bam |           |
| 2024030... B DemoWES_GEM_T_12_GEM_T_12_L00_outfastp_R_deduped.bam |           |
| 2024030... B DemoWES_GEM_T_13_GEM_T_13_L00_outfastp_R_deduped.bam |           |

☐ Make Paths Relative to Project

Check whether the BAM file is aligned to its respective VCF.

Import Variants Wizard

### Import Variant Sources

- Define Input
- Scan Input
- Change Options
- Review

The new source file will be saved to the project data folder.

(Advanced) You may also choose to left align, or split variants into allelic primitives.

☐ Advanced Options

**Summary:**

- Total size: 482M , 66 Files
- 35 fields, Sample Relationship: Unrelated samples(66 affected).
- Assembly: GRCh\_38,Chromosome,Homo sapiens

**Warning:**

**Import Options:**

☐ Subset to Track

☐ Force Call Variants

☒ Use Default Chromosome Names

☐ Subset to Regions

☐ Liftover

☒ Allelic Primitives

☒ Left Align

☐ Variant/CNV Deletion Threshold

☐ Filter Flags

☒ Multi-Allelic Split

Activities \*CNV\_GENEMIND\_REF\_66\_JULY2024 - Golden Helix VarSeq 2.6.1 Jul 23 12:32

\*CNV\_GENEMIND\_REF\_66\_JULY2024 - Golden Helix VarSeq 2.6.1

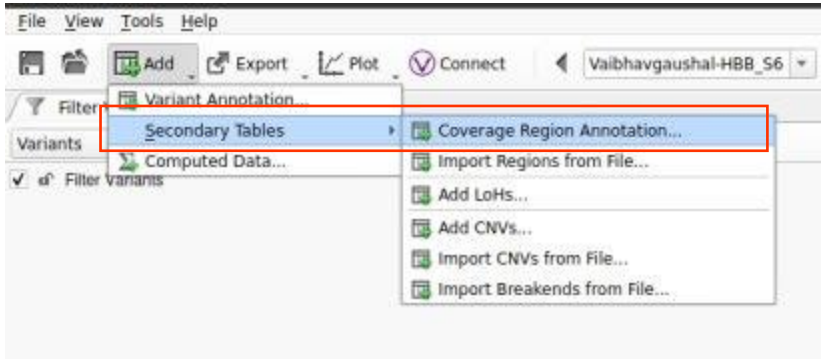
File View Tools Help

202403071719\_AB01231201002\_2P240102010U52S2699DX\_B\_DemoWES\_GEM\_T\_10\_GEM\_T\_10\_L00\_outfastp\_R

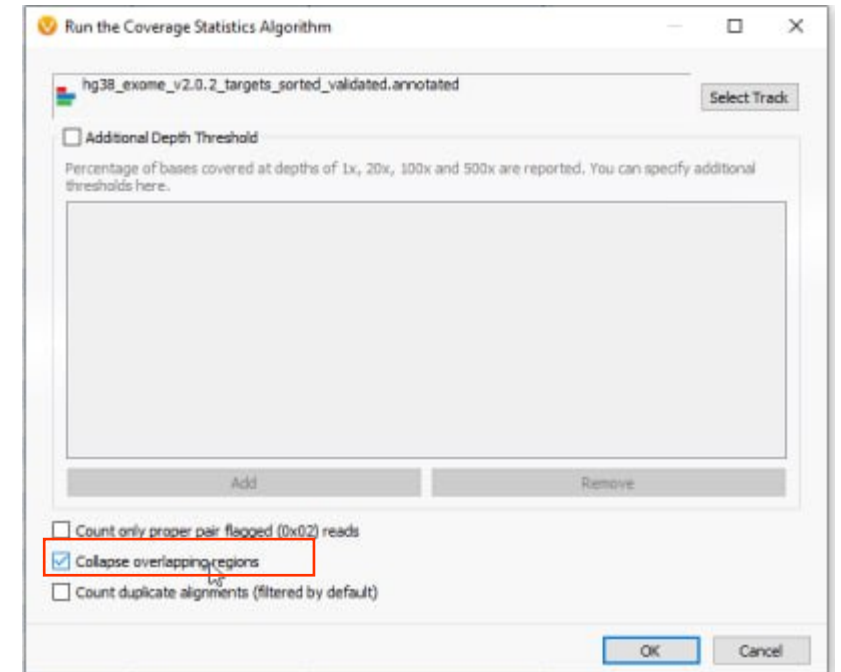
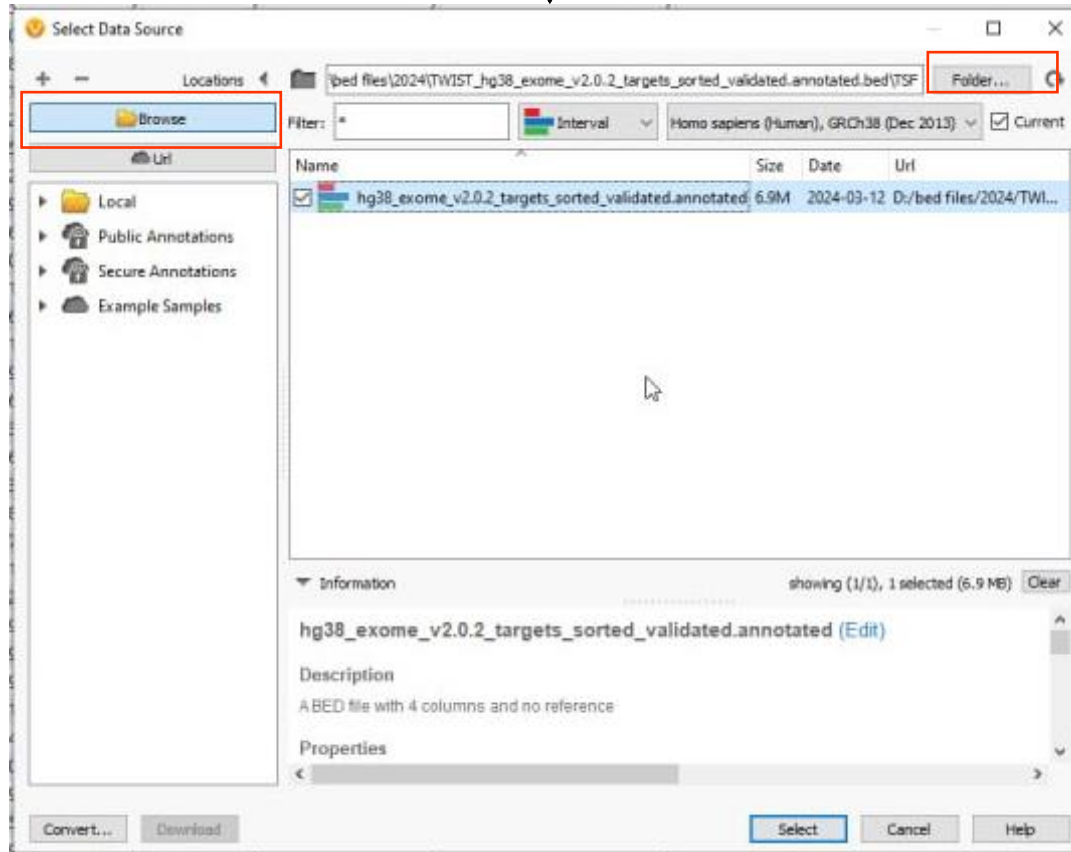
Filter Variants 5,055,541

Filter Variants: 202403071719\_AB01231201002\_2P240102010U52S2699DX\_B\_DemoWES\_GEM\_T\_10\_GEM\_T\_10\_L00\_outfastp\_R

| Chr:Pos | Ref/Alt    | Identifier | Filter  | Variant Allele Fraction | Allelic Depths (AD) | Read Depths (DP) | Genotype Qualities (GQ) | 0/1 Genotypes (GT) |
|---------|------------|------------|---------|-------------------------|---------------------|------------------|-------------------------|--------------------|
| 1:12807 | C/T        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:13273 | G/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:13290 | CT/-       | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:13418 | -/GAGA     | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:13649 | G/C        | ?          | ?       | 1                       | 0,5                 | 5                | 15                      | 1/1                |
| 1:13868 | A/G        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:14464 | A/T        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:15820 | G/T        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:15904 | -/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:15957 | G/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:16020 | C/G        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:16141 | C/T        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:16210 | C/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:16487 | T/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:16495 | G/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:28563 | A/G        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:28591 | -/TGG      | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:30923 | G/T        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:30990 | G/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:51803 | T/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:52238 | T/G        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:54844 | G/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:57856 | T/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:61987 | A/G        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:61989 | G/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:63309 | G/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:63664 | C/T        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:63671 | G/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:63697 | T/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:63736 | CTA/-      | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:69063 | T/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:69270 | A/G        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:69511 | A/G        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:69552 | G/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:69594 | T/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:69849 | G/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:69897 | T/C        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:76838 | T/G        | ?          | LowQual | 1                       | 0,2                 | 2                | 6                       | 1/1                |
| 1:77175 | AA/-       | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:83830 | AGAAAGA... | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:85112 | C/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:87190 | G/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:87409 | C/T        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |
| 1:89145 | G/A        | ?          | ?       | ?                       | 7,7                 | ?                | ?                       | ?                  |



Browse to the folder that has the TSF file.



Click on collapse overlapping regions

File View Tools Help

Connect VaibhavaGaushal-HBB\_S6\_L001\_R ?

Variant Annotation... Secondary Tables Computed Data

Filter Variants: VaibhavaGaushal-HBB\_S6\_L001\_R

| Variant Info | Ref/Alt | Identifier   | Filter  | Variant Allele Fraction | Allelic Depths (AD) | Read Depths (DP) | Genotype Qualities (GQ) | 0/1 Genotypes (GT) |
|--------------|---------|--------------|---------|-------------------------|---------------------|------------------|-------------------------|--------------------|
| 1:3002335    | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:3002350    | G/A     | rs2742678    | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:3757360    | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:3757367    | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:3757378    | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:3757381    | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:9492422    | G/C     | rs12125657   | LowQual | ?                       | 1                   | 0.1              | 1                       | 3                  |
| 1:9492427    | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:16371923   | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:16371932   | A/C     | rs6604909    | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:17997188   | A/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:25903037   | T/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:28173864   | A/G     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:29830335   | T/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:42464719   | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:42464761   | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:51178692   | A/G     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:56214783   | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:58536417   | A/G     | rs2764683    | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:62480013   | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:62480016   | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:62480106   | G/T     | rs78213555   | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:71683163   | A/G     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:73698875   | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:85272173   | A/G     | ?            | LowQual | ?                       | 1                   | 0.2              | 2                       | 6                  |
| 1:93480632   | C/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:93480634   | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:93480753   | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:93480758   | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:1135148... | G/A     | ?            | LowQual | ?                       | 1                   | 0.1              | 1                       | 3                  |
| 1:1471821... | T/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:1502246... | C/T     | rs1499210... | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:1502246... | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:1502248... | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:1502248... | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:1502248... | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  |
| 1:1509917... | A/C     | rs6664622    | ?       | ?                       | ?                   | ?                | ?                       | ?                  |

Select a Source and Field

Select the VAF field that will be used to identify LoH.

Sources: Variants 202403071719\_AB01231201002\_2P2401

Fields:

| Name                     | Type                  | Doc                                    |
|--------------------------|-----------------------|----------------------------------------|
| Filter                   | [M] Categorical Array | Computed filters on the variants.      |
| Variant Allele Fraction  | [M] Float Array       | Computed Variant Allele Fraction       |
| Allelic Depths (AD)      | [M] Int Array         | Allelic depths for the ref and alt ... |
| Read Depths (DP)         | [M] Int               | Read depth                             |
| Genotype Qualities (GQ)  | [M] Int               | Genotype quality                       |
| 0/1 Genotypes (GT)       | [M] String            | Genotype                               |
| PGT                      | [M] String            | Physical phasing haplotype ...         |
| PID                      | [M] String            | Physical phasing ID information, ...   |
| GT Likelihoods Phred ... | [M] Int Array         | The phred-scaled genotype ...          |

Variants: 202403071719\_AB01231201002\_2P240102010US2S2699DX\_B\_DemoWES\_GEM\_T\_1\_GEM\_T\_1\_L00\_outfastp\_1 [merged with 65 other(s)]

Import Without Appending

Sample Relationship: Unrelated

CNV Genemind Ref 66 Aug 2024 Template with 66 samples.

OK Cancel

VarSeq

Min Variants Considered: 30

Expected LoH Rate: Somewhat Common

Min Variant GQ: 30

OK Cancel Help

No changes, keep the default value.

ADD LOHs will run only if the filter chain contains the Variant Allele Fraction



File View Tools Help

Connect Vaibhavgaushal-HBB\_S6\_L001\_R

Filter Variants: Vaibhavgaushal-HBB\_S6\_L001\_R

Variant Annotation... Secondary Tables Computed Data... Filter Variants

Add Coverage Regions... Import Regions from File... Add LoHs... Add CNVs... Import CNVs from File... Import Breakends from File...

| Vaibhavgaushal-HBB_S6_L001_R |         |              |         |                         |                     |                  |                         |                    |     |
|------------------------------|---------|--------------|---------|-------------------------|---------------------|------------------|-------------------------|--------------------|-----|
| Variant Info                 | Ref/Alt | Identifier   | Filter  | Variant Allele Fraction | Allelic Depths (AD) | Read Depths (DP) | Genotype Qualities (GQ) | 0/1 Genotypes (GT) |     |
|                              | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/G     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/G     | rs2742680    | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/T     | rs2742679    | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | rs2742678    | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/C     | rs12125657   | LowQual | ?                       | 1                   | 0.1              | 1                       | 3                  | 1/1 |
|                              | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/C     | rs6604909    | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | T/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/G     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | T/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/G     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/G     | rs2764683    | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/T     | rs78213555   | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/G     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/G     | ?            | LowQual | ?                       | 1                   | 0.2              | 2                       | 6                  | 1/1 |
|                              | C/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | T/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | ?            | LowQual | ?                       | 1                   | 0.1              | 1                       | 3                  | 1/1 |
|                              | T/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/T     | rs1499210... | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | C/T     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/C     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | G/A     | ?            | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |
|                              | A/C     | rs6664622    | ?       | ?                       | ?                   | ?                | ?                       | ?                  | ?   |

VarSeq

Parameters Advanced

Sensitivity  Precision

More CNVs Balanced Fewer CNVs

Current Selection: (3) Balanced

Reference Samples

Minimum Number of Reference Samples

Maximum Number of Reference Samples

☒ Exclude reference samples with percent difference greater than

☒ Add samples to reference set

☒ Normalize sex chromosomes using only controls with matching sex

Reference Sample Folder:

QC Flag Thresholds

Controls average target mean depth below

Controls variation coefficient above

Targets Excluded From Normalization and CNV Calling

Create a filtered targets track from your reference samples using the Create Low Quality Targets wizard found in the [Tools] > [Manage Reference Samples] dialog.

Select an interval source that defines the filtered targets.

Note: Only one coverage reference samples folder should be there.

Change the Max.no of reference samples to 100 (incase if your reference samples is more than 50).

Orelse, don't change any parameters.

Variants: 5,055,541 X    LoHs: 15,603 X    Coverage Regions: 283,942 X    CNVs: 521 X    **Samples: 66 X** +

Samples

| Coverage Statistics hg38_exome_v2.0.2_targets_sorted_validated.annotated |                            |             |              |               |               |               | Sample Flags | Copy Number Variants |              |                      |                        |                 |                 |                |
|--------------------------------------------------------------------------|----------------------------|-------------|--------------|---------------|---------------|---------------|--------------|----------------------|--------------|----------------------|------------------------|-----------------|-----------------|----------------|
| Sample Mean Reverse ...                                                  | Sample Mean Filtered Depth | Sample % 1x | Sample % 20x | Sample % 100x | Sample % 500x | Sample Errors |              | Inferred Gender      | # CNV Events | # Flagged CNV Events | # Unflagged CNV Events | # Hom Deletions | # Het Deletions | # Duplications |
| 33.3149                                                                  | 1.67922                    | 99.132      | 97.9447      | 13.619        | 0.00990442    | ?             | ?            | Male                 | 1752         | 912                  | 840                    | 91              | 235             | 129            |
| 35.2163                                                                  | 1.77283                    | 99.0737     | 98.1753      | 9.83782       | 0.0101348     | ?             | ?            | Female               | 1400         | 679                  | 721                    | 87              | 160             | 105            |
| 33.851                                                                   | 1.5525                     | 99.0529     | 98           | 9.09599       | 0.00771019    | ?             | ?            | Female               | 1500         | 737                  | 763                    | 91              | 167             | 112            |
| 41.793                                                                   | 1.9841                     | 99.0528     | 98.3599      | 26.481        | 0.0109193     | ?             | ?            | Female               | 1503         | 683                  | 820                    | 79              | 173             | 110            |
| 40.5406                                                                  | 1.86458                    | 99.1834     | 98.3851      | 24.7453       | 0.0113417     | ?             | ?            | Male                 | 1520         | 736                  | 784                    | 76              | 193             | 110            |
| 34.4396                                                                  | 1.62866                    | 99.0584     | 98.0584      | 9.55871       | 0.00570787    | ?             | ?            | Female               | 1364         | 656                  | 708                    | 74              | 183             | 100            |
| 34.9908                                                                  | 1.66418                    | 99.1953     | 98.0632      | 11.3464       | 0.00744136    | ?             | ?            | Male                 | 1571         | 761                  | 810                    | 82              | 217             | 114            |
| 34.2583                                                                  | 1.43485                    | 99.189      | 97.8825      | 11.9182       | 0.00355747    | ?             | ?            | Male                 | 1818         | 831                  | 987                    | 78              | 226             | 135            |
| 51.5125                                                                  | 2.24522                    | 99.105      | 98.4468      | 50.2152       | 0.0144575     | ?             | ?            | Male                 | 1182         | 669                  | 513                    | 90              | 129             | 83             |
| 41.5718                                                                  | 1.85394                    | 99.1167     | 98.1901      | 29.618        | 0.0107931     | ?             | ?            | Male                 | 1288         | 719                  | 569                    | 101             | 164             | 89             |
| 47.9647                                                                  | 2.21702                    | 99.0793     | 98.299       | 43.4938       | 0.010442      | ?             | ?            | Male                 | 1333         | 782                  | 551                    | 117             | 225             | 80             |
| 58.4068                                                                  | 2.55912                    | 99.201      | 98.5872      | 60.2147       | 0.0217809     | ?             | ?            | Male                 | 1417         | 658                  | 759                    | 69              | 184             | 100            |
| 46.3105                                                                  | 2.033                      | 99.1077     | 98.3538      | 39.8111       | 0.00934757    | ?             | ?            | Male                 | 1572         | 817                  | 755                    | 94              | 152             | 112            |
| 55.2905                                                                  | 2.56881                    | 98.9441     | 98.3663      | 56.9045       | 0.013259      | ?             | ?            | Female               | 1081         | 671                  | 410                    | 120             | 131             | 68             |
| 44.171                                                                   | 2.01153                    | 99.0926     | 98.2655      | 34.6983       | 0.00771282    | ?             | ?            | Male                 | 1084         | 689                  | 395                    | 100             | 177             | 70             |
| 76.529                                                                   | 3.80884                    | 98.8825     | 98.4077      | 83.7633       | 0.0308268     | ?             | ?            | Male                 | 1038         | 656                  | 382                    | 222             | 398             | 31             |
| 80.8855                                                                  | 4.47105                    | 98.8408     | 98.3024      | 83.3655       | 0.035536      | ?             | ?            | Male                 | 1237         | 698                  | 539                    | 219             | 683             | 25             |
| 72.2702                                                                  | 3.76716                    | 98.715      | 98.2476      | 79.3735       | 0.0178531     | ?             | ?            | Female               | 1032         | 670                  | 362                    | 280             | 445             | 22             |
| 54.5259                                                                  | 2.9597                     | 98.7278     | 97.9262      | 53.6292       | 0.00779236    | ?             | ?            | Male                 | 1643         | 1050                 | 593                    | 331             | 939             | 31             |
| 80.4617                                                                  | 4.05761                    | 98.9317     | 98.4655      | 86.3529       | 0.0352702     | ?             | ?            | Male                 | 869          | 588                  | 281                    | 197             | 285             | 25             |
| 78.3613                                                                  | 3.97841                    | 98.8877     | 98.3403      | 84.9103       | 0.0356432     | ?             | ?            | Female               | 848          | 541                  | 307                    | 165             | 362             | 24             |
| 76.3105                                                                  | 3.62403                    | 98.9559     | 98.464       | 84.8187       | 0.0295679     | ?             | ?            | Male                 | 930          | 613                  | 317                    | 160             | 262             | 38             |
| 50.6683                                                                  | 3.16001                    | 98.5879     | 97.8096      | 47.1835       | 0.00539787    | ?             | ?            | Female               | 3914         | 1946                 | 1968                   | 522             | 2164            | 118            |
| 66.1304                                                                  | 3.28323                    | 98.8647     | 98.3437      | 74.0182       | 0.0191422     | ?             | ?            | Male                 | 932          | 613                  | 319                    | 254             | 346             | 23             |
| 63.8663                                                                  | 3.16748                    | 98.7773     | 98.2254      | 71.0504       | 0.0171483     | ?             | ?            | Female               | 936          | 658                  | 278                    | 266             | 378             | 18             |
| 81.5086                                                                  | 4.27135                    | 98.8619     | 98.3452      | 84.8336       | 0.0376617     | ?             | ?            | Male                 | 995          | 660                  | 335                    | 213             | 501             | 21             |
| 75.7372                                                                  | 3.69943                    | 98.7951     | 98.2282      | 81.6243       | 0.0254233     | ?             | ?            | Female               | 922          | 691                  | 231                    | 249             | 423             | 15             |
| 107.853                                                                  | 5.44413                    | 98.7992     | 98.4075      | 94.9916       | 0.142368      | ?             | ?            | Female               | 822          | 562                  | 260                    | 232             | 320             | 12             |
| 82.7679                                                                  | 4.25163                    | 98.9051     | 98.4646      | 87.552        | 0.0430954     | ?             | ?            | Male                 | 1017         | 649                  | 368                    | 236             | 351             | 31             |
| 88.0069                                                                  | 3.95509                    | 98.8864     | 98.3416      | 80.8051       | 0.0662641     | ?             | ?            | Female               | 1333         | 907                  | 426                    | 223             | 869             | 17             |
| 66.9024                                                                  | 4.28049                    | 98.6656     | 98.1609      | 73.5112       | 0.0144849     | ?             | ?            | Female               | 3226         | 1850                 | 1376                   | 511             | 1706            | 95             |
| 88.3255                                                                  | 4.48773                    | 98.9334     | 98.4491      | 88.3705       | 0.0449112     | ?             | ?            | Male                 | 823          | 584                  | 239                    | 207             | 378             | 16             |
| 59.716                                                                   | 2.78449                    | 98.796      | 98.0778      | 64.2076       | 0.0106423     | ?             | ?            | Female               | 1061         | 700                  | 361                    | 244             | 479             | 22             |
| 84.823                                                                   | 4.0765                     | 98.8563     | 98.3585      | 88.8274       | 0.041192      | ?             | ?            | Female               | 906          | 585                  | 321                    | 204             | 325             | 20             |
| 99.737                                                                   | 4.90884                    | 98.9048     | 98.4558      | 91.8447       | 0.0778767     | ?             | ?            | Male                 | 772          | 579                  | 193                    | 224             | 342             | 13             |
| 68.0092                                                                  | 3.09983                    | 98.9298     | 98.3333      | 75.0599       | 0.0230481     | ?             | ?            | Male                 | 874          | 644                  | 230                    | 191             | 360             | 23             |
| 87.7094                                                                  | 5.02534                    | 98.8581     | 98.4259      | 88.7435       | 0.0412468     | ?             | ?            | Male                 | 1990         | 1491                 | 499                    | 388             | 1236            | 28             |
| 72.2978                                                                  | 3.83474                    | 98.7419     | 98.2303      | 70.0424       | 0.0594444     | ?             | ?            | Male                 | 1298         | 874                  | 424                    | 344             | 555             | 27             |
| 64.3078                                                                  | 3.2146                     | 98.8541     | 98.1796      | 63.4304       | 0.020969      | ?             | ?            | Male                 | 966          | 713                  | 253                    | 283             | 412             | 18             |

go to the samples table, and check the sample flags, (Check the CNV caller tutorial in Golden Helix web page to know what sample flags to remove)



# Building the workflow

Filter CNVs 44,811

Filter CNVs: VIGNESHRAJAM\_A\_RUN\_03\_A\_23\_L00\_outfastp\_R... 167

| Filter                       | CNV Info           | Flags | Avg Target Mean Depth | Avg Z Score | Avg Ratio | Estimated CN | Supporting LOH Variants | Karyotype | GC Content | p-value              | Gene List   |
|------------------------------|--------------------|-------|-----------------------|-------------|-----------|--------------|-------------------------|-----------|------------|----------------------|-------------|
| ✓ CNV State (Current) is (C) | 128 Het Deletion   | ?     | 47.72                 | -2.17773    | 0.570688  | 1            | 0                       | ?         | 0.664062   | 8.48125532447952e... | SAMD11      |
|                              | 409 Het Deletion   | ?     | 41.748                | -2.29455    | 0.554889  | 1            | 0                       | ?         | 0.665037   | 3.41967574414156e... | ACAP3       |
|                              | 4368 Duplicate     | ?     | 212.178               | 1.75423     | 1.64638   | 3            | 1                       | ?         | 0.551593   | 0.000145622367513... | ?           |
|                              | 49 Het Deletion    | ?     | 48                    | -2.73059    | 0.367177  | 1            | 0                       | ?         | 0.367347   | 0.000523448910653... | CASZ1       |
|                              | 160 Het Deletion   | ?     | 25                    | -1.99924    | 0.546628  | 1            | 0                       | ?         | 0.525      | 2.92942603392302e... | PLOD1       |
|                              | 177 Het Deletion   | ?     | 26.7259               | -1.9493     | 0.520651  | 1            | 0                       | ?         | 0.638418   | 1.5552214292876e-05  | ATP13A2     |
|                              | 268 Het Deletion   | ?     | 39.272                | -1.95703    | 0.605287  | 1            | 0                       | ?         | 0.675373   | 1.43793582240885e... | HSPG2       |
|                              | 349 Duplicate      | ?     | 157.592               | 3.25353     | 2.0214    | 4            | 0                       | ?         | 0.633238   | 5.98633689855906e... | CELA3B      |
|                              | 139 Het Deletion   | ?     | 51.5                  | -1.55271    | 0.596538  | 1            | 1                       | ?         | 0.597122   | 6.9469318185876e-05  | SELENON     |
|                              | 76 Het Deletion    | ?     | 32.8851               | -2.29886    | 0.450553  | 1            | 0                       | ?         | 0.631579   | 3.30419612571638e... | CSF3R       |
|                              | 89074 Duplicate    | ?     | 228.997               | 1.01668     | 1.37765   | 3            | 2                       | ?         | 0.35667    | 9.40044790780054e... | AMY1A,AM... |
|                              | 2152 Duplicate     | ?     | 89.8913               | 1.67085     | 1.70442   | 3            | 0                       | ?         | 0.466543   | 0.000304208018121... | PDE4DIP     |
|                              | 221 Duplicate      | ?     | 67.5                  | 1.44567     | 1.31213   | 3            | 0                       | ?         | 0.475113   | 9.46452220659677e... | IRF6        |
|                              | 204 Duplicate      | ?     | 168.811               | 3.28215     | 1.44554   | 3            | 0                       | ?         | 0.431373   | 4.38223804972795e... | USH2A       |
|                              | 74996 Duplicate    | ?     | 155.688               | 1.69168     | 1.24354   | 3            | 0                       | ?         | 0.359779   | 1.51058654022101e... | USH2A       |
|                              | 4673 Duplicate     | ?     | 104.74                | 1.29393     | 1.26667   | 3            | 0                       | ?         | 0.365932   | 7.77190843774048e... | USH2A       |
|                              | 15632 Duplicate    | ?     | 245.639               | 2.61364     | 1.35941   | 3            | 0                       | ?         | 0.45183    | 1.46540845263171e... | GALM,SR5F7  |
|                              | 176 Duplicate      | ?     | 173.8                 | 1.90745     | 1.30934   | 3            | 0                       | ?         | 0.5        | 8.07605072776412e... | ABCG8       |
|                              | 112 Duplicate      | ?     | 144.75                | 2.26391     | 1.33756   | 3            | 0                       | ?         | 0.348214   | 1.14006656746961e... | TMEM127     |
|                              | 43518 Het Deletion | ?     | 73.3778               | -2.92553    | 0.461539  | 1            | 0                       | ?         | 0.398594   | 3.67043012574095e... | RGPD3       |
|                              | 90469 Duplicate    | ?     | 211.926               | 1.6587      | 1.29963   | 3            | 2                       | ?         | 0.418541   | 3.51621006635745e... | LIMS1       |
|                              | 31 Duplicate       | ?     | 94.6667               | 2.00329     | 1.41527   | 3            | 0                       | ?         | 0.516129   | 1.33067318743414e... | IL1RN       |
|                              | 57 Het Deletion    | ?     | 56                    | -2.68013    | 0.597248  | 1            | 0                       | ?         | 0.54386    | 6.1967070053013e-12  | PAXB        |
|                              | 1188 Duplicate     | ?     | 238.826               | 1.51625     | 1.54595   | 3            | 0                       | ?         | 0.367003   | 0.000154260937679... | NEB         |
|                              | 19430 Het Deletion | ?     | 50.75                 | -1.71629    | 0.575327  | 1            | 0                       | ?         | 0.379156   | 1.08287130286264e... | COL3A1,C... |
|                              | 196 Duplicate      | ?     | 162.615               | 1.69154     | 1.2928    | 3            | 1                       | ?         | 0.44898    | 1.85317803153142e... | BARD1       |
|                              | 26492 Duplicate    | ?     | 237.86                | 1.60739     | 1.3182    | 3            | 0                       | ?         | 0.361921   | 1.11829752387061e... | MFF         |
|                              | 372 Duplicate      | ?     | 153.214               | 1.97039     | 1.32722   | 3            | 0                       | ?         | 0.408602   | 1.2896816456091e-20  | NDUFA10     |
|                              | 209 Duplicate      | ?     | 118.4                 | 1.74491     | 1.3164    | 3            | 0                       | ?         | 0.483254   | 9.66291990620235e... | COLQ        |
|                              | 133 Duplicate      | ?     | 84.6889               | 1.60995     | 1.29235   | 3            | 0                       | ?         | 0.526316   | 5.45146192467569e... | ?           |
|                              | 35538 Duplicate    | ?     | 247.288               | 2.0426      | 1.35429   | 3            | 1                       | ?         | 0.382914   | 9.34036876581488e... | CNTN3       |
|                              | 86 Het Deletion    | ?     | 50.75                 | -1.75065    | 0.567427  | 1            | 0                       | ?         | 0.313953   | 7.17539595673964e... | UMPS        |
|                              | 279 Het Deletion   | ?     | 46.5                  | -1.75652    | 0.448724  | 1            | 0                       | ?         | 0.318996   | 6.68276903331321e... | CP,HP53     |
|                              | 44 Duplicate       | ?     | 238.091               | 2.9276      | 1.66875   | 3            | 0                       | ?         | 0.545455   | 0.000239275811824... | NMD3        |
|                              | 155 Duplicate      | ?     | 185.803               | 2.22326     | 1.42613   | 3            | 0                       | ?         | 0.354839   | 8.04811030847965e... | SI          |
|                              | 124 Duplicate      | ?     | 158.703               | 2.60033     | 1.38737   | 3            | 0                       | ?         | 0.435484   | 1.3127138021354e-08  | SEP5ECS     |
|                              | 426 Duplicate      | ?     | 121.682               | 1.58736     | 1.30998   | 3            | 0                       | ?         | 0.488263   | 3.57022073362076e... | CCT5        |
|                              | 154 Duplicate      | ?     | 95                    | 1.62301     | 1.40934   | 3            | 0                       | ?         | 0.337662   | 0.000457589801315... | IL7R        |
|                              | 7420 Duplicate     | ?     | 90.3595               | 1.99694     | 3.34765   | 4            | 0                       | ?         | 0.375741   | 5.09717711348993e... | GTF2H2C     |
|                              | 85108 Deletion     | ?     | 22.4374               | -0.662755   | 0.227166  | 0            | 0                       | ?         | 0.422945   | 0.00302205401193418  | GTF2H2,N... |
|                              | 26307 Het Deletion | ?     | 87.0108               | -2.39348    | 0.580746  | 1            | 1                       | ?         | 0.423368   | 6.40842683676087e... | PCDH11,B... |

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Add the below to filter chain

- \* CNV state
- \* Filter flags
- \* p-Value - 0.05

Variants: 5,055,541 | LoHs: 15,603 | Coverage Regions: 283,942 | CNVs: 521 | Samples: 66

Filter CNVs: 202403071719\_ABO1231201002\_2P240102010US2526990X\_B\_DemoWES\_GEM\_T\_2\_GEM\_T\_2\_L00\_outfastp\_R... #8

Region | Type | # Targets

|                 |      |    |
|-----------------|------|----|
| 1:1041478-10... | Gain | 5  |
| 1:10630161-1... | Gain | 2  |
| 1:12774936-1... | Gain | 62 |
| 1:16124370-1... | Gain | 4  |
| 1:22081722-2... | Gain | 3  |
| 1:39511174-3... | Gain | 6  |
| 1:42733603-4... | Gain | 9  |
| 1:45332966-4... | Gain | 15 |
| 1:55056233-5... | Gain | 9  |
| 1:55063763-5... | Gain | 11 |
| 1:55064387-5... | Loss | 4  |
| 1:89010904-8... | Loss | 3  |
| 1:89058841-8... | Gain | 3  |
| 1:92247878-9... | Gain | 7  |
| 1:113836548...  | Loss | 1  |
| 1:120763628...  | Loss | 5  |
| 1:145643869...  | Gain | 11 |
| 1:143672730...  | Loss | 1  |
| 1:143678449...  | Gain | 4  |
| 1:146974739...  | Gain | 5  |
| 1:148533861...  | Gain | 3  |
| 1:148607083...  | Gain | 2  |
| 1:148962428...  | Gain | 1  |
| 1:149008444...  | Gain | 2  |
| 1:149390788...  | Gain | 4  |
| 1:149923542...  | Gain | 3  |
| 1:150321489...  | Loss | 4  |
| 1:155610425...  | Gain | 6  |
| 1:166922372...  | Gain | 3  |
| 1:179509296...  | Gain | 3  |
| 1:179551460...  | Gain | 2  |
| 1:196741868...  | Gain | 26 |
| 1:233808444...  | Gain | 2  |
| 1:236902941...  | Gain | 5  |
| 1:237831505...  | Gain | 5  |
| 1:237833025...  | Gain | 13 |
| 1:248441117...  | Gain | 4  |
| 2:38982074-3... | Gain | 54 |
| 2:43852347-4... | Gain | 5  |

Select Data Source

Select tracks to use as annotation sources against the imported variant set.

Locations: Local

Filter: (Any type) | Homo sapiens (Human), GRCh38 (Dec 2013) | Current

| Name                                                                                 | Type     | Size    | Date       | Url                      |
|--------------------------------------------------------------------------------------|----------|---------|------------|--------------------------|
| <input checked="" type="checkbox"/> 1kG Phase3 CNVs and Large Variants 5b V2, GHI    | Interval | 2.4M    | 2020-05-13 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> 1kG Phase3 CNVs and Large Variants 5b V2, GHI               | Interval | 2.4M    | 2020-05-13 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> ACMG_Fusion_Case_Catalog_GRCh_38                            | Interval | Unknown | 2024-05-24 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> ACMG_Fusion_Case_Catalog_GRCh_38                            | Interval | Unknown | 2024-05-24 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> Anderson_ACMG Classifier Germline Variant                   | Interval | Unknown | 2023-08-17 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> Cancer Ontology 2023-03-01                                  | Tabular  | 148K    | 2023-03-01 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> ClinGen Gene Disease Validity 2024-02-06, NCBI              | Interval | 1.3M    | 2024-02-06 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> ClinGen Gene Dosage Sensitivity 2024-07-04, NCBI            | Interval | 768K    | 2024-07-04 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> ClinGen Region Dosage Sensitivity 2024-07-04, NCBI          | Interval | 208K    | 2024-07-04 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> Clinical Genomic Database 2023-12-01, GHI                   | Interval | 824K    | 2024-01-02 | C:/Users/NGSWINDOWS/A... |
| <input checked="" type="checkbox"/> ClinVar CNVs and Large Variants 2024-07-04, NCBI | Interval | 6.7M    | 2024-07-04 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> ClinVar Transcript Counts 2024-07-04, NCBI                  | Interval | 1.8M    | 2024-07-07 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> ClinVarCNVsandLargeVariant Assessments 2024-04-04, NCBI     | Interval | 6.0M    | 2024-04-04 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> Conservation Scores Exonic, GHI                             | Value    | 1.2G    | 2020-10-30 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> Cytobands 2014-06-11, UCSC                                  | Cytoband | 88K     | 2016-06-10 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> DECIPHER Developmental Disorders 2024-03-01, GHI            | Interval | 416K    | 2024-03-01 | C:/Users/NGSWINDOWS/A... |
| <input type="checkbox"/> DECIPHER Population CNV v9.2                                | Interval | 2.2M    | 2019-12-16 | C:/Users/NGSWINDOWS/A... |

Information showing (51/251), 2 selected (9.1 MB) Clear

ClinVar CNVs and Large Variants 2024-07-04, NCBI (Edit)

Description

The ClinVar database from NCBI contains information about the phenotypes and supporting evidence for variants in the dbSNP database. For more information about the ClinVar database please see: [About ClinVar](#)

Properties

Type Interval

Convert... Download Select Cancel Help

Add 1Kg phase 3 CNV  
and ClinVar CNV db



Filter V X Filt X Filt X + Variants: 252,322 X Coverage Regions: 283,942 X LoHs: 319 X CNVs: 226 X Sample: 1 X +

CNVs

Filter CNVs 1,770

✓ # Matches Type <= 3 OR missing

# Matches Type - Summary of CNVs 1...

3

Less than 3 182

Equal to 3 1

Greater than 3 1

Missing 0

183

✓ Clinical Significance (per Condition) is

Clinical Significance (per Condition) - ...

Affects 0

Association 0

Benign 8

Benign; Likely Benign 0

Conflicting Data from Submitters 1

Conflicting Interpretations of Pathogenicity 0

Drug Response 0

Likely Benign 5

Likely Pathogenic 4

Likely Pathogenic; Low Penetrance 0

No Classifications from Unflagged Records 0

Not Provided 4

Other 0

Pathogenic 3

Pathogenic; Likely Pathogenic 1

Pathogenic; Low Penetrance 0

Protective 0

Risk Factor 0

Uncertain Risk Allele 0

Uncertain Significance 11

Uncertain Significance; Pathogenic; Likely Path 0

Overlapping Transcripts RefSeq Genes 110, NCBI

Summary of CNVs 1kG Phase3 CNVs and Large Variants 5b ...

| HGVS c.   | HGVS p.       | Sequence Ontology                          | Gene Region                   | # Matched | # Gains | # Losses | # Matches Type | Region       | Span |
|-----------|---------------|--------------------------------------------|-------------------------------|-----------|---------|----------|----------------|--------------|------|
| 03352...  | NP_27707...   | frameshift_variant,frameshift_variant,...  | exon,exon,exon,exon           | 0         | 0       | 0        | 0              | ?            |      |
| 03352...  | NP_27707...   | inframe_deletion,inframe_deletion,infr...  | exon,exon,exon,exon           | 0         | 0       | 0        | 0              | ?            |      |
| 00101...  | NP_00101...   | frameshift_variant                         | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| 01794...  | ?             | transcript_ablation                        | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| ?         | ?             | ?                                          | ?                             | 0         | 0       | 0        | 0              | ?            |      |
| 03081...  | NP_11044...   | disruptive_inframe_deletion                | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| ?,?,?,?,? | ?,?,?,?,?     | unknown,unknown,unknown,unknown...         | unknown,unknown,unkn...       | 0         | 0       | 0        | 0              | ?            |      |
| 00136...  | ?,?,?,?,?,?   | intron_variant,intron_variant,intron_va... | intron,intron,intron,intro... | 0         | 0       | 0        | 0              | ?            |      |
| ?,?,?     | ?,?,?         | unknown,unknown,unknown                    | unknown,unknown,unkn...       | 0         | 0       | 0        | 0              | ?            |      |
| ?,?,?     | ?,?,?         | unknown,unknown,unknown                    | unknown,unknown,unkn...       | 0         | 0       | 0        | 0              | ?            |      |
| 00128...  | NP_00126...   | frameshift_variant,frameshift_variant,...  | exon,exon,exon                | 0         | 0       | 0        | 0              | ?            |      |
| 00134...  | NP_00133...   | initiator_codon_variant,initiator_codon... | exon,exon,exon,intron,e...    | 0         | 0       | 0        | 0              | ?            |      |
| 00120...  | ?,NP_0026...  | transcript_ablation,inframe_deletion,i...  | exon,exon,exon,exon,exon      | 0         | 0       | 0        | 0              | ?            |      |
| 01538...  | NP_05619...   | initiator_codon_variant,initiator_codon... | exon,exon,unknown,unk...      | 0         | 0       | 0        | 0              | ?            |      |
| ?,?       | ?,?           | unknown,unknown                            | unknown,unknown               | 0         | 0       | 0        | 0              | ?            |      |
| 00110...  | NP_00109...   | frameshift_variant,frameshift_variant      | exon,exon                     | 0         | 0       | 0        | 0              | ?            |      |
| 20693...  | ?,?           | intron_variant,intron_variant              | intron,intron                 | 0         | 0       | 0        | 0              | ?            |      |
| 00128...  | NP_00127...   | frameshift_variant,frameshift_variant,...  | exon,exon,exon,exon           | 0         | 0       | 0        | 0              | ?            |      |
| ?         | ?             | unknown                                    | unknown                       | 0         | 0       | 0        | 0              | ?            |      |
| ?         | ?             | unknown                                    | unknown                       | 0         | 0       | 0        | 0              | ?            |      |
| 00111...  | ?,?,?         | intron_variant,intron_variant,intron_va... | intron,intron,intron          | 0         | 0       | 0        | 0              | ?            |      |
| 00102...  | ?,?,?         | transcript_ablation,transcript_ablation... | exon,exon,unknown             | 0         | 0       | 0        | 0              | ?            |      |
| 1_0010... | ?,NP_0010...  | unknown,stop_lost,stop_lost                | unknown,exon,exon             | 0         | 0       | 0        | 0              | ?            |      |
| 00107...  | NP_00107...   | inframe_deletion,inframe_deletion          | exon,exon                     | 0         | 0       | 0        | 0              | ?            |      |
| ?         | ?             | ?                                          | ?                             | 0         | 0       | 0        | 0              | ?            |      |
| 00131...  | NP_00129...   | inframe_deletion,inframe_deletion          | exon,exon                     | 1         | 0       | 1        | 1              | 2:9589209... | 1701 |
| 18258...  | NP_87239...   | inframe_deletion                           | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| 00116...  | NP_00115...   | frameshift_variant                         | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| ?         | ?             | ?                                          | ?                             | 0         | 0       | 0        | 0              | ?            |      |
| 00130...  | NP_00129...   | inframe_deletion,inframe_deletion,infr...  | exon,exon,exon,exon,ex...     | 0         | 0       | 0        | 0              | ?            |      |
| 02518...  | ?,?,NP_001... | intron_variant,intron_variant,disruptiv... | intron,intron,exon,intron     | 0         | 0       | 0        | 0              | ?            |      |
| 00109...  | NP_00109...   | inframe_deletion                           | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| 20731...  | NP_99719...   | initiator_codon_variant                    | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| 00127...  | NP_00126...   | inframe_deletion,inframe_deletion          | exon,exon                     | 0         | 0       | 0        | 0              | ?            |      |
| 00108...  | ?             | transcript_ablation                        | exon                          | 0         | 0       | 0        | 0              | ?            |      |
| ?,?       | ?,?           | unknown,unknown                            | unknown,unknown               | 0         | 0       | 0        | 0              | ?            |      |
| 00454...  | ?,?,?,?       | intron_variant,transcript_ablation,tran... | intron,exon,exon,exon         | 0         | 0       | 0        | 0              | ?            |      |
| ?,?,?,?,? | ?,?,?,?,?,?   | unknown,unknown,unknown,unknown...         | unknown,unknown,unkn...       | 0         | 0       | 0        | 0              | ?            |      |
| 00692...  | ?,?,?,?,?,?   | intron_variant,intron_variant,intron_va... | intron,intron,intron,intro... | 0         | 0       | 0        | 0              | ?            |      |
| 13343...  | ?,?,?,?,?,?   | intron_variant,intron_variant,intron_va... | intron,intron,intron,intro... | 0         | 0       | 0        | 0              | ?            |      |
| 02453...  | ?             | intron_variant                             | intron                        | 0         | 0       | 0        | 0              | ?            |      |
| 00110...  | ?             | intron_variant                             | intron                        | 0         | 0       | 0        | 0              | ?            |      |
| ?         | ?             | unknown                                    | unknown                       | 0         | 0       | 0        | 0              | ?            |      |

Add **Matched** to the filter chain - Keep the value **3**.







File View Tools Help

GEN\_T\_2.0.2\_0063\_outfastp\_R

Filter V x Filter x Filter x +

Filter CNVs 1,770

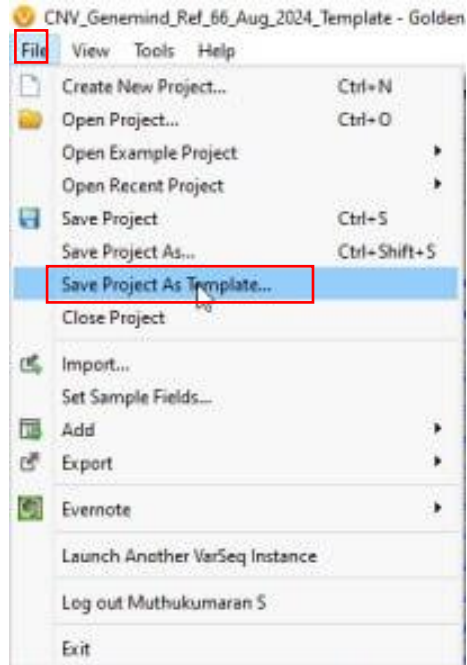
ACMG CNV Site Classifier

| Gene ...  | Critical Gene List | Scored Gene | Scored Gene Transcript | Scored Gene Impact Score | Additional Score | Total Score | Classification | Criteria       | Potential Gene Imp... |
|-----------|--------------------|-------------|------------------------|--------------------------|------------------|-------------|----------------|----------------|-----------------------|
| CDK11A    | ?                  | CDK11A      | NM_024011.4            | 0                        | -1               | -1          | Benign         | 2B:0 While...  |                       |
| CDK11A    | ?                  | CDK11A      | NM_024011.4            | 0                        | -1               | -1          | Benign         | 2B:0 While...  |                       |
| PRAMEF7   | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| NBPF1     | ?                  | ?           | ?                      | ?                        | 0                | 0           | VUS            | 1A:0 The C...  |                       |
| ?         | ?                  | ?           | ?                      | ?                        | -1.6             | -1.6        | Benign         | 1B:-0.6 Th...  |                       |
| IKRD13C   | ?                  | ?           | ?                      | ?                        | 0                | 0           | VUS            | 1A:0 The C...  |                       |
| ACADM     | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| ADGRL2    | ?                  | ADGRL2      | NM_001366006.2         | -0.45                    | 0                | -0.45       | VUS            | 2H:0.15 Th...  |                       |
| DBT       | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| DBT       | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| SYCP1     | ?                  | ?           | ?                      | ?                        | 0                | 0           | VUS            | 1A:0 The C...  |                       |
| FAM72C    | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| PDZK1     | ?                  | ?           | ?                      | ?                        | 0                | 0           | VUS            | 1A:0 The C...  |                       |
| PF14,N... | ?                  | ?           | ?                      | ?                        | 0                | 0           | VUS            | 1A:0 The C...  | 0                     |
| SLC19A2   | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| SWT1      | ?                  | ?           | ?                      | ?                        | 0                | 0           | VUS            | 1A:0 The C...  |                       |
| USH2A     | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| MARK1     | ?                  | ?           | ?                      | ?                        | 0                | 0           | VUS            | 1A:0 The C...  |                       |
| KLF11     | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| MATN3     | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| FANCL     | ?                  | ?           | ?                      | ?                        | -1.6             | -1.6        | Benign         | 1B:-0.6 Th...  |                       |
| GLB1,R... | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| GLB2,R... | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| RGPD2     | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| ?         | ?                  | ?           | ?                      | ?                        | -1.6             | -1.6        | Benign         | 1B:-0.6 Th...  |                       |
| IKRD36C   | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| RGPD4     | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| RGPD8     | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| ?         | ?                  | ?           | ?                      | ?                        | -1.6             | -1.6        | Benign         | 1B:-0.6 Th...  |                       |
| RABL2A    | RABL2A             | RABL2A      | NM_001306158.2         | -1                       | -1               | -2          | Benign         | 2F:-1 The c... |                       |
| SLC35F5   | ?                  | ?           | ?                      | ?                        | -1.6             | -1.6        | Benign         | 1B:-0.6 Th...  |                       |
| POTEF     | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| TUBA3E    | ?                  | TUBA3E      | NM_207312.3            | 0                        | -1               | -1          | Benign         | 2F:-1 The c... |                       |
| POTEF     | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| POTEE     | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| ZEB2      | ZEB2               | ZEB2        | NM_014795.4            | -0.6                     | 0                | -0.6        | VUS            | 2D:1:0 The...  |                       |
| NEB       | ?                  | ?           | ?                      | ?                        | -1               | -1          | Benign         | 1A:0 The C...  |                       |
| ACVR1     | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| SCN1A     | SCN1A              | SCN1A       | NM_001165963.4         | -0.6                     | 0                | -0.6        | VUS            | 2B:0 The c...  |                       |
| TTN       | TTN                | TTN         | NM_001267550.2         | -0.6                     | 0                | -0.6        | VUS            | 2B:0 The c...  |                       |
| SPAG16    | ?                  | ?           | ?                      | ?                        | -0.6             | -0.6        | VUS            | 1B:-0.6 Th...  |                       |
| AQP12B    | ?                  | AQP12B      | NM_001102467.2         | -0.6                     | 0                | -0.6        | VUS            | 2B:0 While...  |                       |
| SFMBT1    | ?                  | ?           | ?                      | ?                        | -1.6             | -1.6        | Benign         | 1B:-0.6 Th...  |                       |

Classification - ACMG CNV Site Classi...

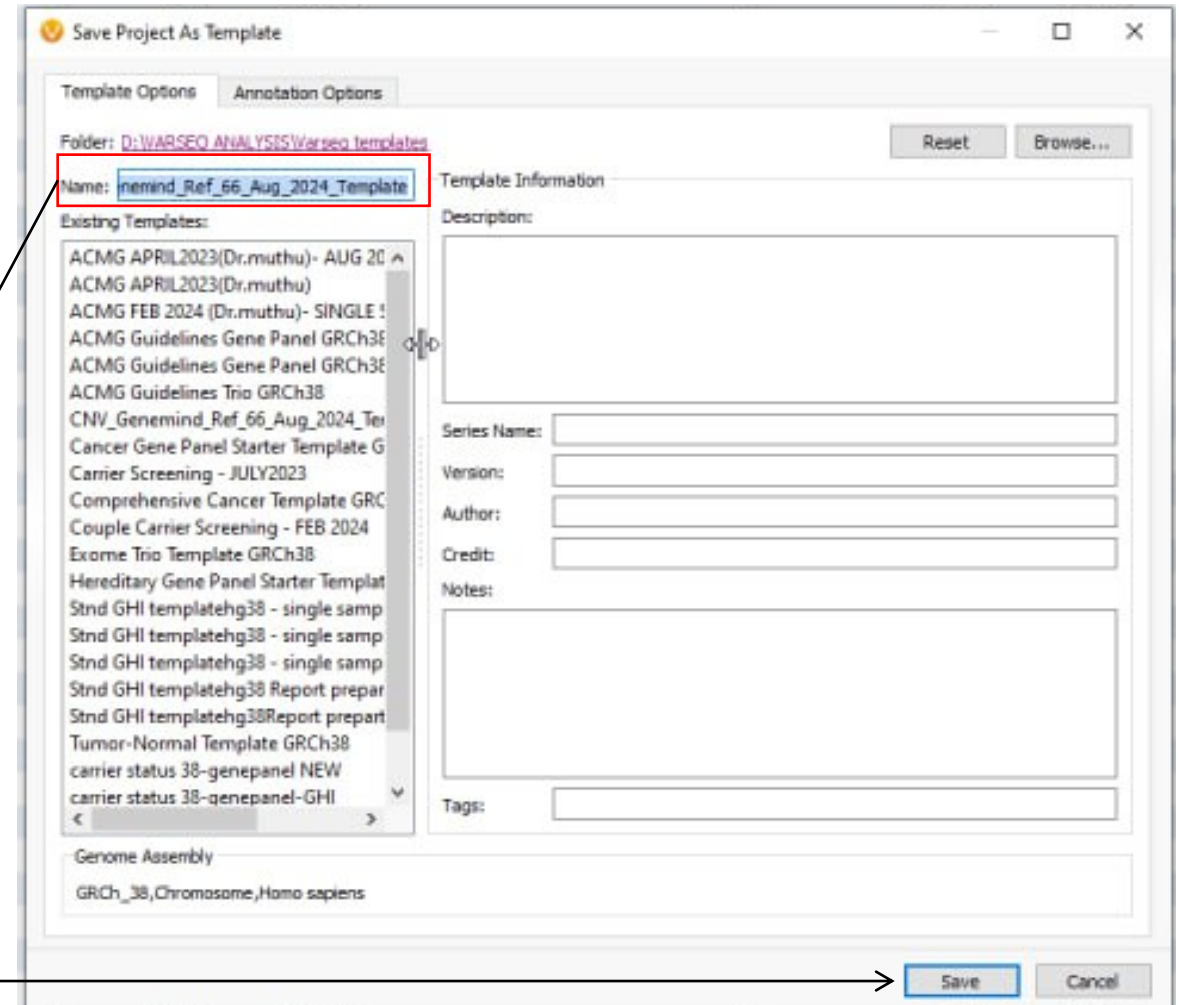
Add  
Classification  
from ACMG  
CNV Site  
Classifier to  
the filter chain.

# Template Creation



Go to **File**,  
Click **Save  
Project as  
Template**

Rename the file and  
save it.





# CNV Analysis

- To do the CNV analysis, Load the VCF file and **choose** --- **Append samples with same names and Variants, CNV** - and BAM in the project template created.
- After loading, check for updates.
- Go to **Add** --- **Computed Data** --- **Table type** - **CNV** ---- **CNV PhoRank Gene Ranking**.
- Leave the default option and click OK.
- Go to **Input sources** --- **Select PhoRank Research (Ontology Propagation)**
- Come back to **Phenotype Selection** --- Type the phenotypes and give ok.
- After loading, add **Gene Rank** to **Filter chain** and keep it above **ClinVar**.
- **Note: Beware of X linked CNVs - High chance of false call.**

Import Variants Wizard

### Import Variant Sources

1 Define Input  
2 Scan Input  
3 Change Options  
4 Review

Select one or more variant files to import.

Select Files:

Add Files  
Remove  
Add Folder  
Remove All

Sample Name Matching Mode

☒ Import Without Appending  
☐ Append Samples with the Same Names  
☐ Append Samples with Similar Names (Shared Prefixes)

Import Tables

☒ Variant Only  
☐ Variants, CNVs  
☐ Variants, CNVs, Breakends

☐ Advanced Options

Help

< Back Next > Cancel

Click on this while loading the VCF file

Select an Algorithm

Pick an algorithm to run on the project data

Algorithms:

Table type: CNVs

- Variant
  - Per Sample
    - Compute Fields
    - Annotate Overlapping CNVs Matching Sample
    - Annotate Overlapping Regions Matching Sample
    - ACMG Sample CNV Classifier
  - Project/Cohort
    - Latest CNV Sample Assessments
    - Annotate Cytobands
    - Annotate Overlapping Genes
    - Annotate Overlapping Regions
    - Annotate Overlapping CNVs
    - Aggregate Compute Fields
    - CNV PhoRank Gene Ranking**
    - Copy Number Probability/Segregation
    - ACMG CNV Site Classifier
  - Gene
    - Per Sample
      - Match Genes List (Per Sample)
      - Match Panels (Per Sample)
      - CNV Sample PhoRank Gene Ranking
    - Project/Cohort
      - Match Genes Linked to Disorders
      - Match Genes Linked to Phenotypes
      - Match Genes and Panels
      - Match String List

Documentation

Algorithm: **CNV PhoRank Gene Ranking** Version: 1

## 4.11.8. CNV PhoRank Gene Ranking

This algorithm ranks CNVs and genes based on their relevance to user-specified phenotypes as defined by the HPO and GO biomedical ontologies. There are two versions of the algorithm: PhoRank Clinical and PhoRank Research.

### PhoRank Clinical

PhoRank Clinical is based on an algorithm published by [Masino et al.](#), that orders genes by the semantic similarity between the phenotypes associated with each gene and those associated with the patient. This method works well when applied to individuals presenting with disease phenotypes that have established gene associations.

### PhoRank Research

PhoRank Research is modeled on the [Phevor](#) algorithm. Phevor assigns scores to ontology terms based on their proximity to the user-specified phenotypes. Nodes that are connected to a search term, either directly or through a shared gene relationship, are called seed nodes and are assigned an initial score of one. The algorithm propagates this score information through the ontologies, so that genes with high scores are more closely related to the specified phenotypes, while genes with low scores have little or no relation to the

Cancel OK

To add Gene Rank

Select a Source and Field

Select the Gene Names field for PhoRank to use for ranking.

Sources

- Copy Number Variants
- Summary of Overlapping hg38\_exome\_
- Summary of CNVs ClinVar CNVs and Lar
- Summary of CNVs ClinVarCNVsandLarge
- Copy Number Probability/Segregation
- Overlapping Genes RefSeq Genes 110,
- Summary of CNVs 1kG Phase3 CNVs an

Fields

Filter \*

| Name                    | Type              | Doc                           |
|-------------------------|-------------------|-------------------------------|
| Gene Names              | String Array      | The set of unique gene ...    |
| Gene IDs                | String Array      | The set of Entrez gene ...    |
| Aliases                 | String Array      | Previous names for the se...  |
| # Genes                 | int               | The number of genes ...       |
| Transcript Name ...     | String Array      | The transcript determined...  |
| Overlapping Exons ...   | String Array      | Which exons of the ...        |
| % CDS Covered ...       | Float Array       | The percentage of the exo...  |
| % Covered (Clinicall... | Float Array       | The percentage of the ...     |
| HGV5 g.                 | String            | The associated HGV5 ...       |
| HGV5 c. (Clinically ... | String Array      | The associated HGV5 ...       |
| HGV5 p. (Clinically ... | String Array      | The associated HGV5 ...       |
| ISCN Cytoband ...       | String            | ISCN notation describing t... |
| Sequence Ontology...    | Categorical Array | The sequence ontology ...     |

**Overlapping Genes RefSeq Genes 110, NCBI**

A summary of the genes overlapping the region.

**Description**

A summary of the genes overlapping the region.

**Algorithm:** overlapgene  
**Algorithm Version:** 2  
**Created:** 2024-08-02

Cancel OK

Leave the default options, Click Ok

PhoRank

Phenotype Selection Input Sources

Human Phenotype Ontology:

Human Phenotype Ontology 2024-04-26 Select Track

Scoring Algorithm:

☐ PhoRank Clinical (Symantic Similarity)
 ☒ PhoRank Research (Ontology Propagation)

Gene Ontology:

Gene Ontology 2022-09-21 Select Track

☐ Enhance with OMIM phenotypes

OMIM:

OMIM Phenotype Ontology 2022-09-21 Select Track

Cancel OK

Select the second option

PhoRank

Phenotype Selection Input Sources

Phenotype Group Name:

Enter a comma separated list of phenotype terms...

Cancel OK

Type the phenotypes and give ok